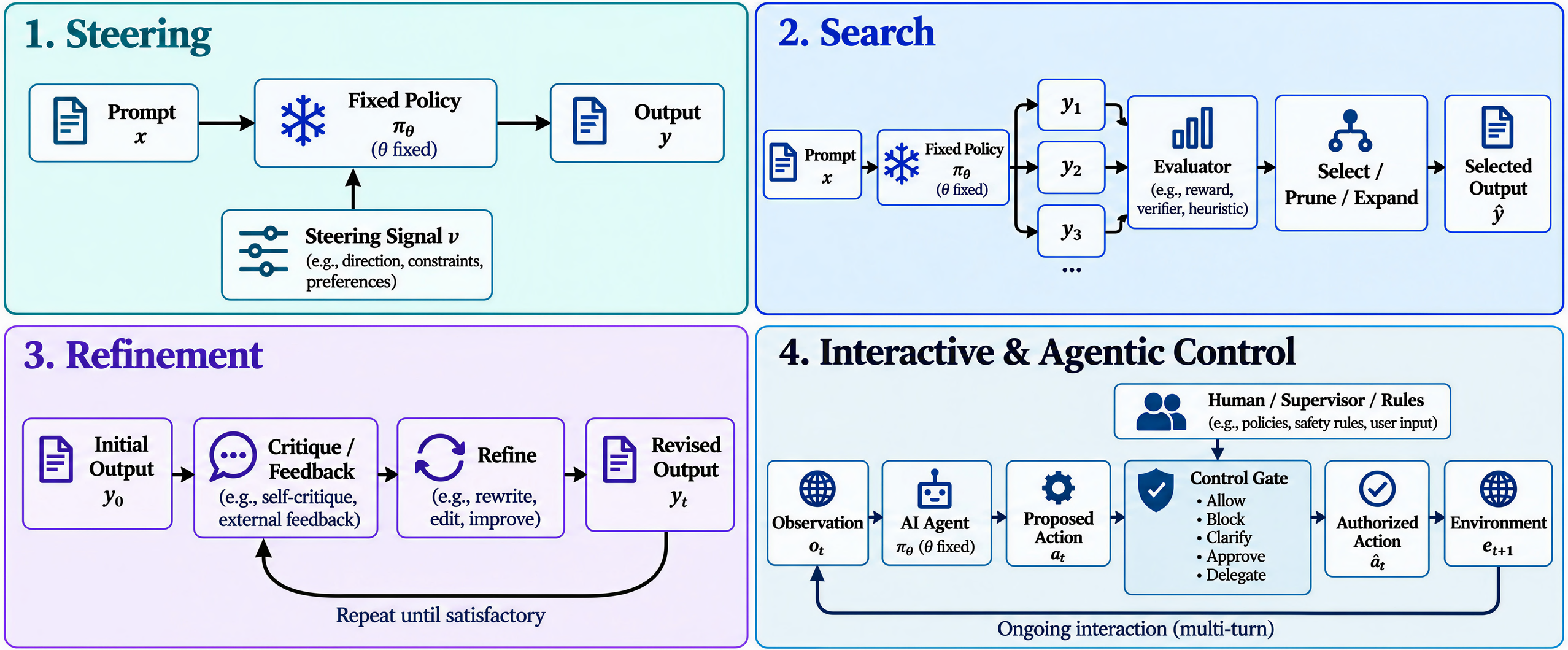

1. Steering
Prompt
x
Fixed Policy
π_θ
(θ fixed)
Output
y
Steering Signal v
(e.g., direction, constraints, preferences)
2. Search
Prompt
x
Fixed Policy
π_θ
(θ fixed)
y_1
y_2
y_3
...
Evaluator
(e.g., reward, verifier, heuristic)
Select / Prune / Expand
Selected Output
$\hat{y}$
3. Refinement
Initial Output
y_0
Critique / Feedback
(e.g., self-critique, external feedback)
Refine
(e.g., rewrite, edit, improve)
Revised Output
y_t
Repeat until satisfactory
4. Interactive & Agentic Control
Human / Supervisor / Rules
(e.g., policies, safety rules, user input)
Observation
o_t
AI Agent
π_θ (θ fixed)
Proposed Action
a_t
Control Gate
• Allow
• Block
• Clarify
• Approve
• Delegate
Authorized Action
$\hat{a}_t$
Environment
e_{t+1}
Ongoing interaction (multi-turn)